

Linear Transformations

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1 Definition

Definition 1.1. $f : E \rightarrow E, x \mapsto f(x)$

f is a Linear Transformation $\iff \forall x, y \in E, \forall \lambda \in \mathbb{R} : f(\lambda x + y) = \lambda f(x) + f(y)$

We denote the set of linear applications as: $\mathcal{L}(E, F)$ and $\mathcal{L}(E, F)E$ if $f : E \rightarrow E$

$$f \text{ is a homomorphism} \iff f \text{ is LT} \tag{1.1}$$

$$f \text{ is an endomorphism} \iff (??) \text{ and } f : E \rightarrow E \tag{1.2}$$

$$f \text{ is an isomorphism} \iff (??) \text{ and } f \text{ is bijective} \tag{1.3}$$

$$f \text{ is an automorphism} \iff (??) \text{ and } (??) \text{ and } (??) \tag{1.4}$$

Properties 1.2. f is a linear transformation, then:

- $f(0_E) = 0_F$
- $f(-x) = -f(x)$

2 Kernel, Image and morphisms

Remark 2.1. f is a linear transformation in the following:

Definition 2.2. (Kernel) $\ker f = \{x \in E \mid f(x) = 0_E\}$

Theorem 2.3. f is injective $\iff \ker f = \{0_E\} \iff \dim(\ker f) = 0$

Definition 2.4. (Image) $\text{Im} f = \{f(x) \mid x \in E\}$

Theorem 2.5. f is surjective $\iff \text{Im} f = F \iff \dim(\text{Im} f) = \dim F$

Theorem 2.6. f is bijective $\iff f$ is injective and surjective.

Properties 2.7. $\dim E = \dim \ker f + \dim \text{Im} f$

Theorem 2.8. If $E = F$ then:

f is injective $\iff f$ is surjective $\iff f$ is bijective $\iff \ker f = \{0_E\} \iff \text{Im} f = E \iff \dim \ker f = 0 \iff \dim \text{Im} f = \dim E$

3 Matrices associated with linear transformation

Definition 3.1. Let $B = \{e_1, e_2, \dots, e_n\}$ a basis of E and $B' = \{e'_1, e'_2, \dots, e'_p\}$ a basis of F , Then the associated matrix of the LT f is given by:

$$Mat = \begin{matrix} & f(e_1) & f(e_2) & \cdots & f(e_n) \\ \begin{matrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{matrix} & \begin{pmatrix} \alpha_1 & \alpha'_1 & \cdots & \alpha_1^{(p)} \\ \alpha_2 & \alpha'_2 & \cdots & \alpha_2^{(p)} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n & \alpha'_n & \cdots & \alpha_n^{(p)} \end{pmatrix} \end{matrix}$$

Remark 3.2. Writing $f(e_k)$ as $f(e_k) = \alpha_1^{(k)} e'_1 + \alpha_2^{(k)} e'_2 + \cdots + \alpha_n^{(k)} e'_n$ then write the coefs in columns ($\alpha_n^{(k)}$ means α k-times prime)

We denote $Mat(f)_{B_1, B_2}$ the matrix associated with the linear application $f : E \rightarrow F$ with the basis B_1, B_2 of E, F respectively. If $f : E \rightarrow E$ then we just denote it $Mat(f)_{B_1}$ or $Mat(f)$.

Remark 3.3. f^n DOES NOT MEAN $f \times f \times \dots \times f$ n- times.

It means $f \circ f \circ \dots \circ f$ (Composition of f with itself n-times)

That composition becomes a matrix power Prop. ??

Properties 3.4. $\forall f, g \in \mathcal{L}(E, F), \forall \lambda \in \mathbb{K}$

1. $Mat(\lambda f) = \lambda Mat(f)$
2. $Mat(f + g) = Mat(f) + Mat(g)$
3. $Mat(g \circ f) = Mat(g) \times Mat(f)$
4. $Mat(f^{-1}) = Mat(f)^{-1}$ (**very important**)
5. $Mat(f^p) = Mat(f)^p$

4 The change of basis:

Definition 4.1. Given a basis $B_1 = \{e_1, e_2, \dots, e_n\}$ and a basis $B_2 = \{e'_1, e'_2, \dots, e'_n\}$ of the same set E.

To find the transition matrix, you need to write each vector e'_k as a linear composition of the basis B_1

$$e'_k = \alpha_1^{(k)} e_1 + \alpha_2^{(k)} e_2 + \cdots + \alpha_n^{(k)} e_n$$

Then write the coefs in columns:

$$P_{B_1 \rightarrow B_2} = \begin{matrix} & e'_1 & e'_2 & \cdots & e'_n \\ \begin{matrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{matrix} & \begin{pmatrix} \alpha'_1 & \alpha''_1 & \cdots & \alpha_1^{(k)} \\ \alpha'_2 & \alpha''_2 & \cdots & \alpha_2^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha'_n & \alpha''_n & \cdots & \alpha_n^{(k)} \end{pmatrix} \end{matrix}$$

Let $f : E \rightarrow F$ such that $f \in \mathcal{L}(E, F)$ and B_1, B_2 are basis of E and F resp.

To find the linear transformation f in another base, we need to find the matrix associated with it in that matrix with the formula:

$$Mat(f)_{B_2, B'_2} = P_{B_1 \rightarrow B_2}^{-1} \times Mat(f)_{B_1, B'_1} \times P_{B'_1 \rightarrow B'_2}$$

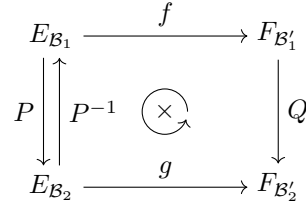


Figure 1: Change of basis Diagram

Such that P is the transformation matrix from $B_1 \rightarrow B_2$ and Q from $B'_1 \rightarrow B'_2$.
 and we write $Mat(f)_{B'_1, B'_2} = P^{-1} \times Mat(f)_{B_1, B_2} \times Q$

Properties 4.2. $f : E \rightarrow E \implies Mat(f)_{B'_1} = P^{-1} \times Mat(f)_{B_1} \times P$

Properties 4.3. $Mat(f)_{B'_1, B'_2}^n = P^{-1} \times Mat(f)_{B_1, B_2}^n \times P$